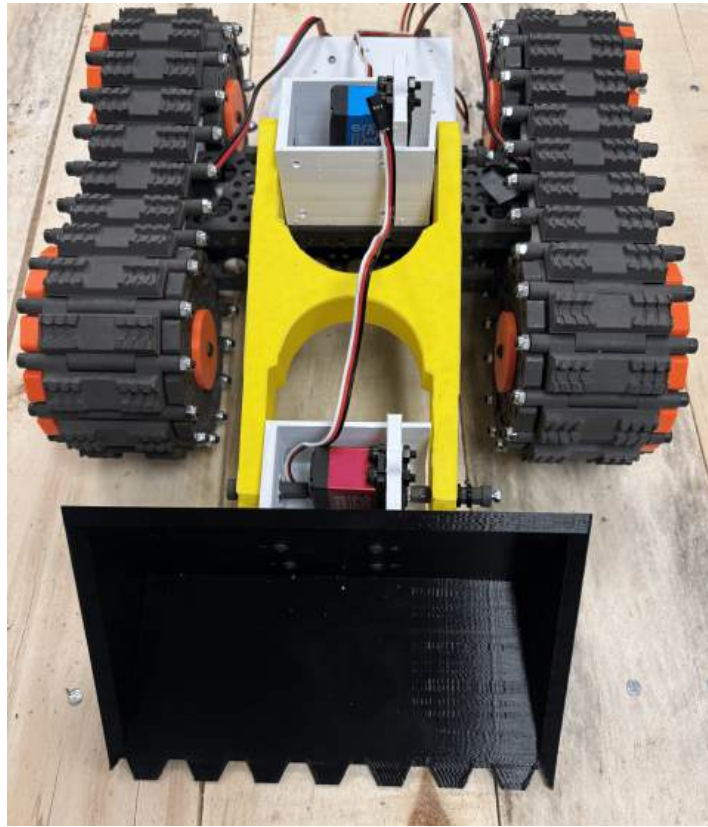


ASME IAM3D Competition Design Report



University of Illinois, Urbana-Champaign
ASME Competitions Committee

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1 Introduction

1.1 Competition overview

This project is developed for the ASME IAM3D Remote Operated Vehicle for the Excavation of Resources (R.O.V.E.R) Competition, which challenges university students to design, manufacture, and test a first-person view (FPV) unmanned vehicle capable of gathering and depositing resources on an extraterrestrial landscape. The R.O.V.E.R must traverse the competition field while successfully picking up, transporting, and releasing resources at a designated point. The competition emphasizes speed, precision, and efficiency, with scoring criteria based on amount of resources deposited and the extent of additive manufacturing integration in the design.

1.2 Project Objective

The goal of this project is to develop a compact and efficient mining robot to extract resources from an extraterrestrial landscape. The R.O.V.E.R must harness the unique capabilities of additive manufacturing to optimize speed, maneuverability, and structural integrity while integrating standard electrical components to minimize costs. In addition, the vehicle must navigate the competition course and collect resources under FPV control to simulate the challenge of remote operations of space vehicles.

1.3 Design Approach

The team employed an iterative design methodology, leveraging computer-aided design (CAD) and rapid prototyping to refine the R.O.V.E.R's performance. Key design considerations included the frame, propulsion system, resource extraction mechanism, remote sensing capabilities, and use of additive manufacturing. The modular frame was optimized for stability, size, and ease of integrating the ancillary components. The propulsion system was selected to maximize traction and maneuverability in the harsh competition landscape, while the resource extraction mechanism was designed to efficiently collect and deposit materials into the resource gathering station.

The frame was optimized for stability, weight reduction, and integration of propulsion components. The propulsion system involved selection of motor count, stator size, and voltage draw along with propeller pitch and length considerations to balance thrust-to-weight ratio and battery longevity for competition. The payload mechanism was designed and iterated over to be an excavation bucket for reliable payload handling. The design is explained in detail in Section 2.

1.4 Expected Outcomes

The final R.O.V.E.R design aims to maximize excavation efficiency, system reliability, and manufacturability. The robot's performance will be evaluated through rigorous testing and validation to ensure optimal performance at the competition. This topic is further elaborated in Section 7.

2 Design Iterations

The team divided the R.O.V.E.R into three mechanical subsystems: the propulsion system, the chassis, and the excavation mechanism. Each system was carefully analyzed using an iterative approach to determine the optimal robot design. These mechanical systems were then integrated with remote control, video, and metal detection systems to build a cohesive R.O.V.E.R.

2.1 Propulsion System

Given the need to quickly traverse a harsh extraterrestrial landscape, the team prioritized an effective and reliable propulsion system. Two options were considered: wheels with rubber tires and a track system.

The wheeled design would significantly reduce complexity and accelerate the development timeline, but the team determined there were significant drawbacks to this option. First, rubber tires could not be additively-manufactured using the fused deposition modeling (FDM) printers available in the student makerspace, thus reducing the additive manufacturing score. Second, tracks offer a significantly

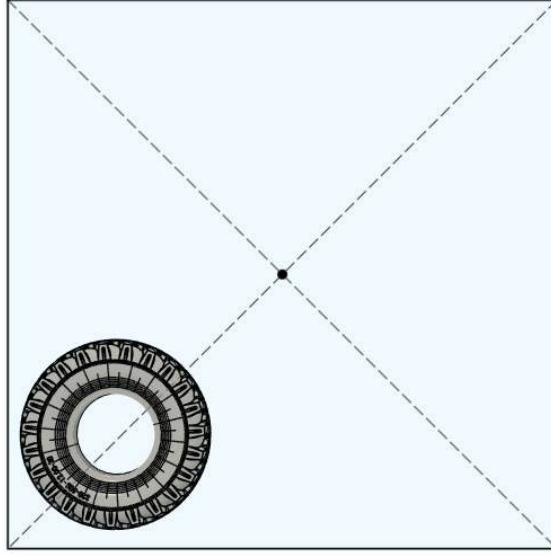


Figure 1: Sample tire in comparison to dimensions of R.O.V.E.R size limit

increased contact area, leading to lower ground pressure and better traction on the uneven granular media of the competition landscape. Finally, the large-volume tires needed to provide sufficient traction occupy a huge portion of the robot's total volume. Figure 1 demonstrates the large area required for the tires in comparison with the 300 mm dimensional sizing limit. Given the strict R.O.V.E.R size requirement, the team determined that tracks were a superior choice.

To maximize the R.O.V.E.R's additive manufacturing score, the team designed a track system around the capabilities of selective-laser-sintering (SLS) printing (which was available for use to the team). Each link is printed from nylon and features three attachment points, two rear and one front, to connect the components into a complete track unit using M4 screws and locknuts. Patterned grooves on the top surface increase the surface area, and thus the traction, of each link. Figure 2 shows a track link.

Each track is tensioned by four PLA wheels which interface with the links at the outside edges to minimize the risk of derailment. The tracks are powered by two 5203 Series Yellow Jacket planetary gear motors with a maximum speed of 223 RPM. This selection provides an appropriate balance of torque and top speed, enabling the R.O.V.E.R to quickly accelerate and traverse across the challenging competition landscape. The motor is connected to two of the wheels through bevel gears and a solid nylon axle; the remaining two wheels spin freely. The axle is attached to the wheels using two bearings and PLA inserts with M4 screws secure the entire assembly. Figure 3 shows the propulsion mechanism with the track removed.

2.2 Chassis

The robot's chassis connects all components of the R.O.V.E.R. Prior to determining the optimal excavation mechanism, the team decided to use a modular chassis design because it simplifies assembly and offers the greatest flexibility for integrating and testing excavation mechanisms.

The team designed three frames shown in Figure 4 for the initial robot chassis. Each design consists of multiple u-shaped bars with hole patterns for flexible mounting positions. The one-piece frames differ in their placement of the horizontal crossbars, offering advantages in electronics packaging and excavation mechanism mounting points.

After receiving input from Dr. Justin Yim, an expert in robotics in UIUC's Novel Mobile Robots Lab, the team determined that sand and dust intrusion into electronic systems posed a significant risk to the robot's reliability. Therefore, the team separated the frame into multiple components and enclosed the drive motors in side frames for improved protection. Positioning the motors inside the side frames also maximizes space utilization, creating additional volume for digging mechanisms mounted on the front of the chassis. Two holes were added on the top of the frames to route the power cable

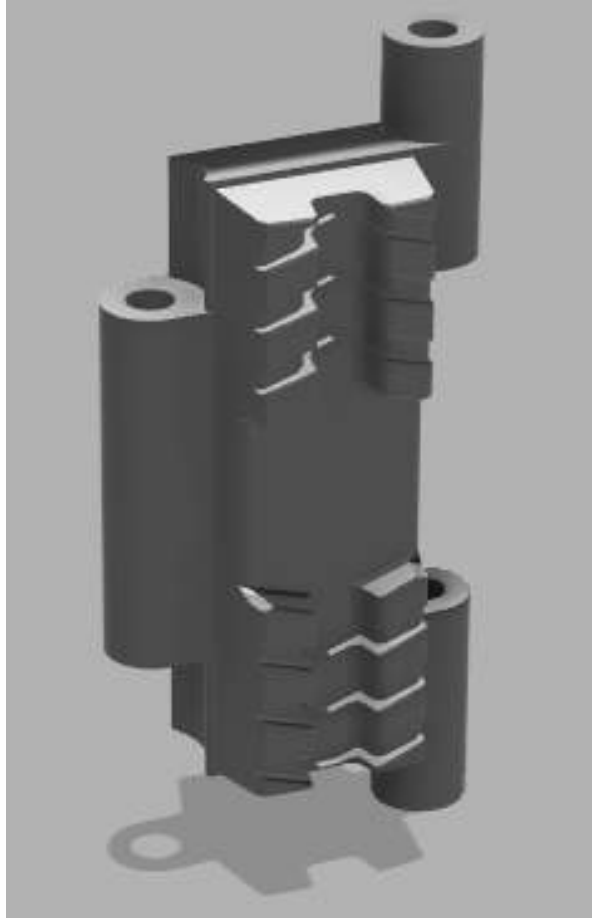
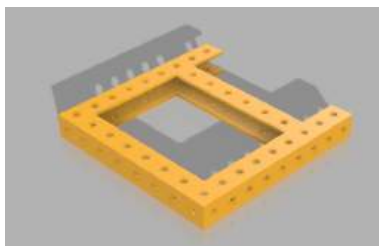


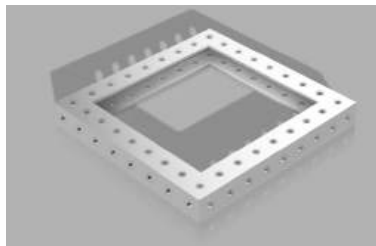
Figure 2: Track link



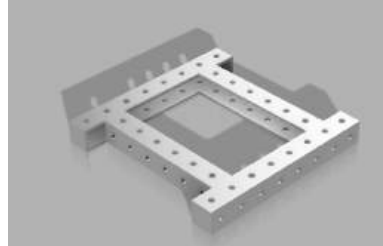
Figure 3: Motor with bevel gears, solid axle, and wheels



(a) Frame 1



(b) Frame 2



(c) Frame 3

Figure 4: Initial designs for R.O.V.E.R chassis

from the drive motor to the microcontroller.

The team iterated through several different side frame designs before arriving at the final model. The first design, shown in Figure 5, was a rectangular prism with one small face removed to install the drive motor. While this approach offered superior protection, the team found it impossible to install interior fasteners to connect to the center frame because an Allen key could not extend far enough from the open face. Additionally, it was difficult to install the drive motor because its position could only be adjusted from a single access point, requiring students to pull on the power cable to remove the motor from the frame.

To correct these issues, the side frame was redesigned as shown in Figure 6. The team removed the remaining small face to allow easier adjustment of the motor's position and included projected holes opposite the fasteners to conveniently insert an Allen Key. These improvements were meaningful, but the team still found it difficult to install interior fasteners. This final issue was solved in the third iteration of the side frame by removing one of the large faces. A cover was added to retain the desired dust protection and semi-circular supports were included to reduce strain on the motor housing. The final frame and cover designs are pictured in Figures 43 and 42 respectively.

The motors are attached to the side frames using a nylon mounting bracket. An M4 thread pattern



Figure 5: Initial design of side frame



Figure 6: Second design of side frame

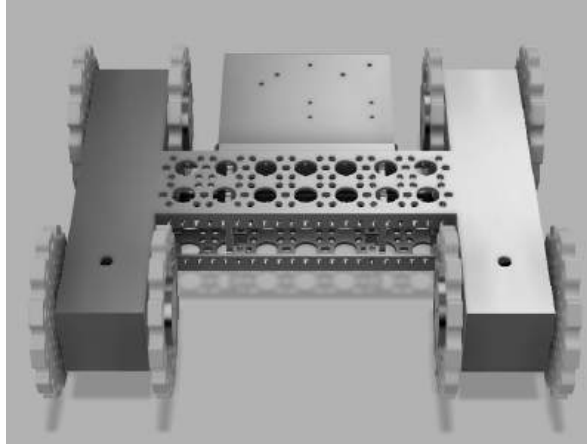


Figure 7: Complete robot without digging mechanism and tracks removed

was created in the bracket by harnessing the highly precise powder fusion of SLS, allowing the team to mount the component directly to the side frame without heat-set inserts.

The center frame consists of two U-shaped beams with four connecting brackets. The center frame and connecting brackets were both printed from nylon to create the same M4 thread pattern found in the drive motor mounting bracket. This design improves assembly efficiency by allowing the team to directly mount the center frame to the connecting brackets without requiring heat-set inserts. The center frame includes hole patterns for flexible mechanism mounting. Additionally, the space inside the center frame is used to house the Li-Po battery powering the R.O.V.E.R.

A triangular PLA bracket was created to mount the control electronics on the rear of the center frame. Figure 7 pictures the complete robot without a digging mechanism installed and the tracks removed for clarity.

2.3 Excavation Mechanism

The team rigorously evaluated five designs of excavation mechanisms based on their size, flexibility, predicted reliability, and complexity to determine the optimal method for extracting resources from the extraterrestrial environment. Three designs, the paddle wheel, rotating scoop, and twin rotating wheels, relied on an auxiliary conveyor belt to deposit resources at the resource gathering station. The excavation wheel and bucket did not require this additional mechanism.

The five mechanisms were evaluated using the Pugh Matrix shown in Table 1. Size refers to the total volume of the digging mechanism. This factor was considered most important given the R.O.V.E.R's strict size limitations. Flexibility refers to the mechanism's ability to mine resources of differing sizes and at different depths. Given that resources range from a 1 cm cube to a 10 cm cube and can be buried up to 10 cm deep, the team regarded this as the second most important design consideration. Reliability was considered the third most important factor because the mechanism must function in a harsh extraterrestrial landscape without direct supervision. Finally, the complexity of each mechanism was compared to determine which designs were most technically feasible.

In the early stages of the design process, the team considered two strategies for gather resources. In the first method, resources would initially be deposited in a container onboard the R.O.V.E.R and then transported to the resource gathering station at the end of the run. In the second method, the resources would be deposited at the resource gathering station as soon as they were identified. Given the R.O.V.E.R's size constraint, the team determined there was not sufficient space on the robot for a preliminary resource storage container. Furthermore, this approach would require an additional mechanism to dump the resources into the final resource station, consuming additional volume and electrical energy. In addition, a storage container would require the chassis to be able to handle additional weight and additionally cause the R.O.V.E.R to be slower in movement. For these reasons, the team elected to design the R.O.V.E.R around method two, depositing resources at the end of the run. This choice proved justified given the robot's performance as described in the testing section.

Criteria	Paddle Wheel (Baseline)	Rotating Scoop	Rotating Wheels	Excavation Wheel	Excavation Bucket
Size	0	-1	-1	-1	0
Flexibility	0	1	0	1	1
Reliability	0	0	-1	-1	1
Complexity	0	0	-1	0	1
Total	0	0	-3	-1	3

Table 1: Pugh Matrix of Excavation Methods

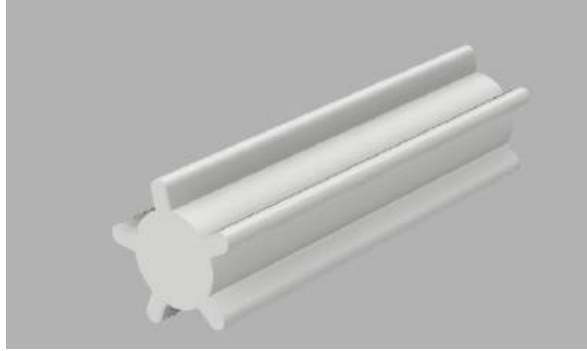


Figure 8: Paddle wheel

2.3.1 Paddle Wheel

The first design pictured in Figure 8 was dubbed the "paddle wheel" because it appears like the mechanisms found on riverboats of the 19th century. This excavation mechanism would rotate in the sand, pushing material just ahead of the robot onto a conveyor belt. The conveyor would diagonally transport resources from the lower front edge to the upper rear edge, depositing them into the resource gathering station. Figure 9 provides a schematic of conveyor belt's functionality. This mechanism was considered the baseline for the Pugh Matrix as it was the first design considered by the team.

2.3.2 Rotating Scoop

The rotating scoop shown in Figure 10 is similar to the paddle wheel, but replaces the five paddles with two curved scoops to improve gathering efficiency. Like the paddle wheel, the scoop would rotate in the sand and push resources onto the conveyor belt, which would deposit them into the resource gathering station. The rotating scoop is slightly larger than the paddle wheel, but enables collection of larger resources, thus scoring equal to the baseline of the Paddle Wheel in the Pugh Matrix.

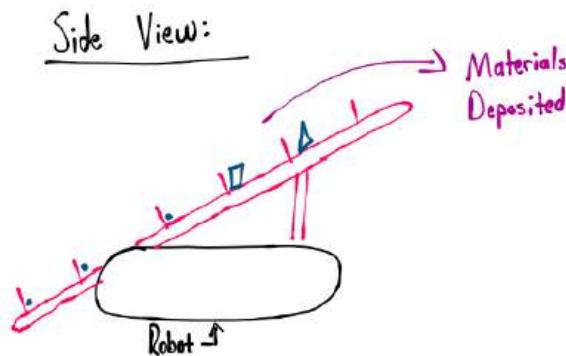


Figure 9: Conveyor belt deposits material into resource gathering station

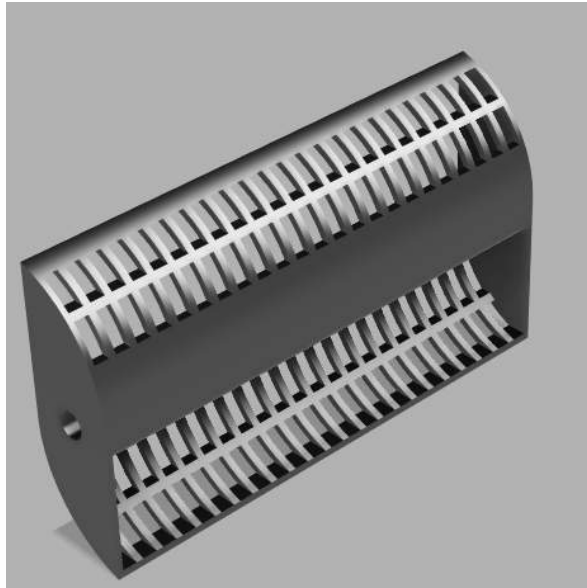


Figure 10: Rotating scoop

2.3.3 Rotating Wheels

The third design shown in Figure 11 uses two side-by-side horizontal wheels to dig through sand and deposit resources onto a conveyor belt. This design is much larger and more complex than the previous two ideas but can handle larger resources more effectively. However, the added complexity and corresponding decrease in predicted reliability do not offset this benefit, making it the least favorable design considered. Additionally, since there are two distinct wheels, there would need to be an additional motor, increasing cost and taking power away from other parts of the R.O.V.E.R.

2.3.4 Excavation Wheel

The excavation wheel pictured in Figure 12 uses five equally-spaced scoops and rotates as it moves through the sand (similar to the previous three ideas). However, its large diameter allows the scoops to reach above the top edge of the resource gathering station and directly deposit resources without requiring the conveyor belt. This simpler design paired with the large size of the scoops makes the design quite flexible for gathering resources of different sizes, but the huge size of the mechanism poses a significant challenge. Additionally, the larger design would require more power, thus straining electrical components and decreasing reliability.

2.3.5 Excavation Bucket

The final design analyzed was a simple bucket shown in Figure 13. This mechanism features a wide opening to gather many resources in one motion and holes to filter out sand. The team reasoned that sand removal would reduce the bucket's weight, reducing energy consumption and stress on components. This design is approximately the same size as the base line while enabling extraction of much larger resources. Additionally, like the excavation wheel, the bucket would not require a conveyor belt system; the bucket can be raised to the height of the resource gathering station and then be tilted so all resources fall into the station. Another advantage was that the bucket would be able to dig through the sand faster than the other designs, improving its flexibility. Furthermore, it is simpler and thus more reliable, making it the best candidate analyzed.

2.3.6 Final Design

After determining the excavation bucket to be the optimal method for extracting resources, the team iterated on the design to improve its performance. Figure 14 shows the modifications to the initial design. First, sharp teeth were added to improve the digging efficiency and the outer edges were

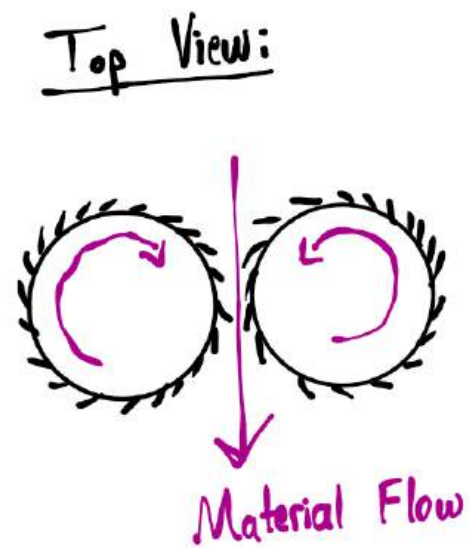


Figure 11: Rotating wheels

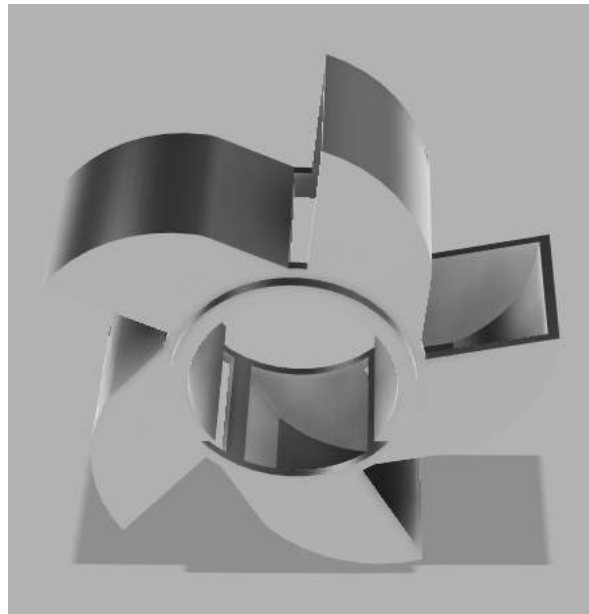


Figure 12: Excavation wheel

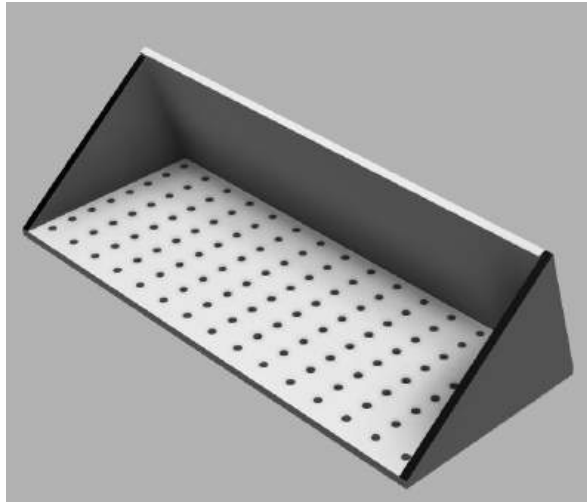


Figure 13: Excavation bucket

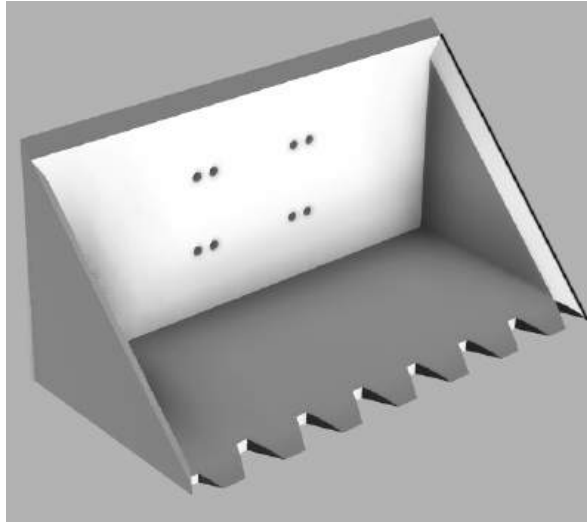


Figure 14: Final design of excavation bucket

tapered to guide resources into the bucket. After consulting with Professor Yim, the team narrowed the bucket's width to distribute the digging force over a smaller area, enabling the mechanism to dig deeper. Finally, a hole pattern was added to the rear face to easily attach the bucket to a servomotor.

Figure 19 shows the complete R.O.V.E.R with the updated extraction bucket and tracks removed for clarity. The bucket is mounted to the top of the center frame using the plough connector: a bracket in the shape of a rounded H.

As with the side frames, the team iterated through several differing designs of the plough connector before arriving at the optimal solution. The first design, shown in Figure 15, was a two piece assembly with identical V-shaped brackets. The team reasoned that the plough would be convenient to replace if one piece of the assembly failed during testing. However, heat-set inserts were required to connect the two halves of the bracket, which greatly increased the assembly time.

To address this issue, the second design, shown in Figure 16 consisted of a single piece bracket with added external reinforcements. Inserts, shown in Figure 17 were added to fasten the reinforcements and adapt to the size of the servo axles. This second iteration also proved time consuming to assemble, but offered better strength over the initial design.

The third design, shown in Figure 18 solved the issue of assembly time by combining the entire plough into a single H-shaped bracket. The hexagonal holes at each corner were adjusted to snugly fit over the servo axles and enlarged the side faces to improve strength. However, when the team



Figure 15: Initial plough connector design



Figure 16: Second plough connector design with external reinforcements

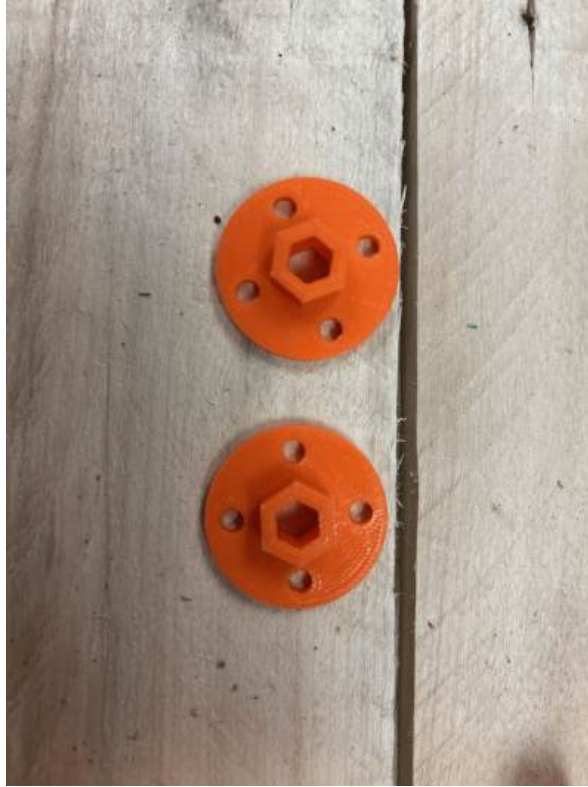


Figure 17: Plough inserts found in second design

tried to install the component, they found it to interfere with the body of the servo housings, making installation impossible.

This issue was solved in the final iteration of the plough connector shown in Figure 39. This design shrank the side faces near the axle connection to remove interference while maintaining rigidity near the horizontal cross-beam.

At each end of the plough connector is a servo housing. Two servomotors produced by SSERVO move the bucket through the entire desired range of motion. The first servo, rated at 35 kg torque, is attached to the center frame and controls the total rotation of the plough connector and bucket. The second servo, rated at 25 kg torque, is on the opposite end of the plough connector and controls solely the rotation of the bucket. A higher torque motor is used on the center frame because of the longer lever arm to the payload in the excavation bucket.

Defining the angle of 0° to be the orientation shown in Figure 19 and taking the positive direction to be counterclockwise when viewing the R.O.V.E.R from the left side, the first servo can rotate from -20° to 150° and the second servo can rotate from -90° to 90° . The expected operation of these motors is further detailed in section 4.

2.4 Remote Control System

The R.O.V.E.R is operated using an Arduino Nano ESP32 microcontroller and an xBox one X controller (paired using a Bluetooth connection). The Bluetooth connection is stable and long-range, ensuring operators always have control over the R.O.V.E.R. The drive motors are operated using BTS 7960 motor drivers. The motor drivers provide sufficient power for R.O.V.E.R to navigate the terrain effectively and quickly. The servomotors are used to maneuver the excavation bucket, quickly moving it from digging to dumping.

2.5 FPV System

The FPV system consists of two ESP32 camera modules with 160° wide angle lenses. The lenses are mounted back to back on the front of the center frame, enabling clear forward and rearward views



Figure 18: Third design of plough connector

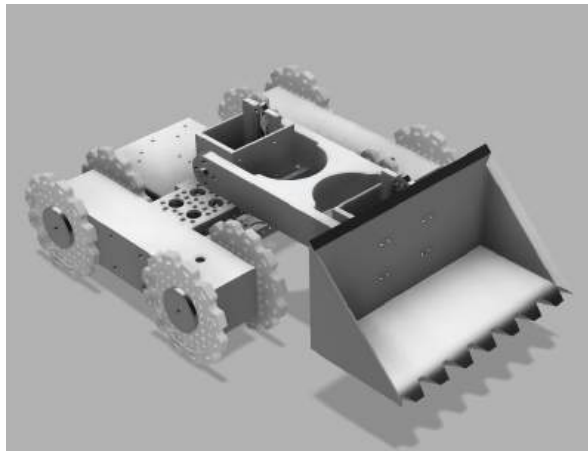


Figure 19: Complete R.O.V.E.R with updated extraction bucket and tracks removed

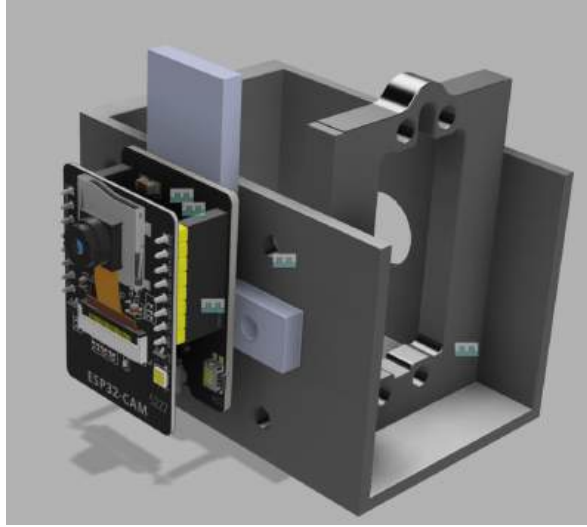


Figure 20: Candidate design for camera mount

from a single mounting location. The team previously considered mounting the cameras to a rotating pedestal for full 360° visibility, but determined this more complex design was unnecessary after testing described in section 7.1. The FPV system connects to the control computer through Wi-Fi, ensuring the ability of the operator to smoothly operate the R.O.V.E.R.

At the time of writing, the mounting assembly for the FPV camera system has not been finalized and was thus not included in drawings. Figure 20 shows a candidate design: adhesive secures the camera and microcontroller to a backboard mounted on the front of the servo housing.

2.6 Metal Detection System

Given that metal objects are worth 1000 points per object retrieved, the team made a concentrated effort to locate and extract metal resources. A Sparkfun linear 3D Hall effect sensor with a range of 80 mT was mounted on the front of the center frame to detect metal object buried in the landscape. When a target is detected, the sensor will output a sound on the ground station laptop and blink a light on the microcontroller, alerting the driver to the presence of the target. This system improves extraction efficiency by locating high-value targets and improves the teams score. Importantly, it will enable the detection of targets even if not visible (if the target is buried), enabling the R.O.V.E.R to obtain even hidden resources.

3 Final Design Drawings

Figure 21 shows a complete assembly drawing of the R.O.V.E.R with tracks removed for clarity. The robot consists of nine principal components as labeled in the parts list. A center frame and connecting brackets serve as the core connecting the side frames, tracks, electronics mounting, and excavation bucket mechanism. Figure 22 shows an exploded assembly drawing of the R.O.V.E.R. For clarity, the track links were removed and only one track assembly was exploded.

Figure 23 shows an exploded view of the track assembly. This assembly contains nine components as detailed in the parts list (the track link is not included). All components mount to the side frame with a top cover completely enclosing the drive motor (not pictured) and bevel gears. The axle assembly consists of a solid axle, bevel gear (on the drive axle), and two bearings, spacers, wheels, and wheel inserts. The entire axle is secured using an M4 screw inserted lengthwise into the axle at each end.

Figures 32 through 46 contain detailed drawings of every component in the R.O.V.E.R. Table 2 summarizes each part with listed quantity, material, and link to the part drawing. With 101 additively-manufactured components and 14 non-additively-manufactured components, the total additive manufacturing score of the R.O.V.E.R is 87.8%.

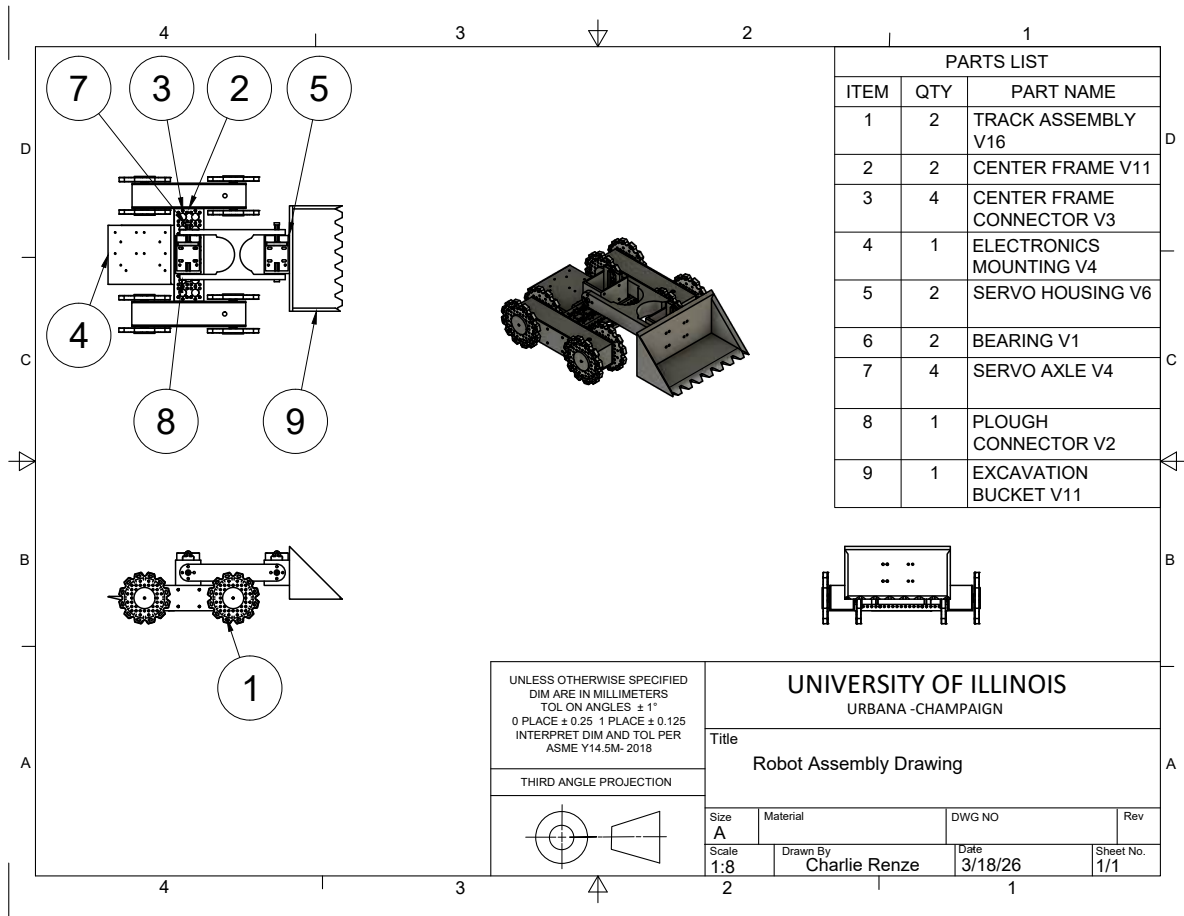


Figure 21: R.O.V.E.R assembly drawing

Component	Quantity	Material	Drawing
Axle Spacer	8	PLA	Figure 32
Center Frame Connector	4	Nylon	Figure 33
Center Frame	2	Nylon	Figure 34
Drive Axle	4	Nylon	Figure 35
Electronics Mounting	1	PLA	Figure 36
Excavation Bucket	1	PLA	Figure 37
Motor Mounting Bracket	2	Nylon	Figure 38
Plough Connector	1	PLA	Figure 39
Servo Housing	2	PLA	Figure 40
Servo Axle	4	Nylon	Figure 41
Side Frame Cover	2	PLA	Figure 42
Side Frame	2	PLA	Figure 43
Track Link	52	Nylon	Figure 44
Track Wheel	8	PLA	Figure 45
Track Wheel Insert	8	PLA	Figure 46
Non-Additively-Manufactured Components			
Bevel Gears	4	N/A	N/A
Bearings	10	N/A	N/A

Table 2: Summary of components in R.O.V.E.R with links to part drawings

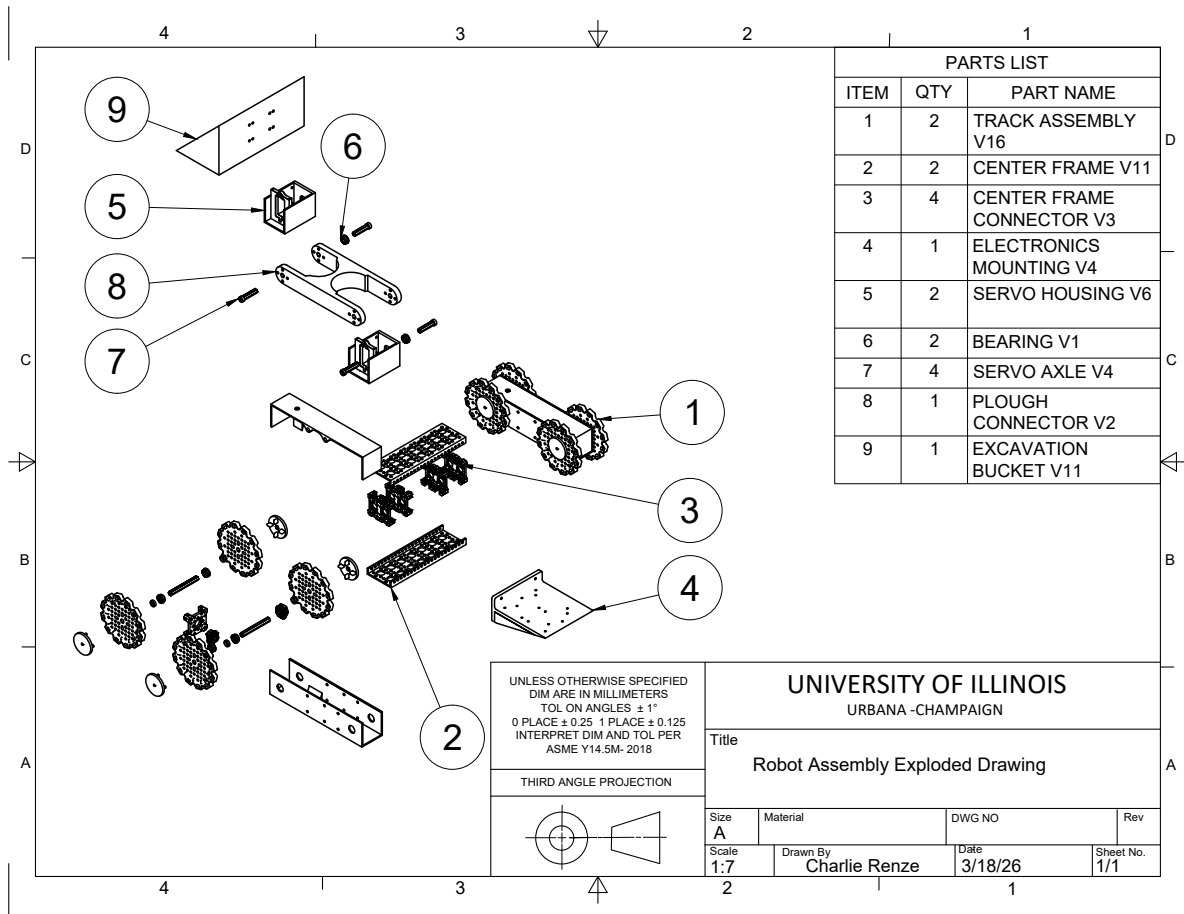


Figure 22: R.O.V.E.R exploded assembly drawing

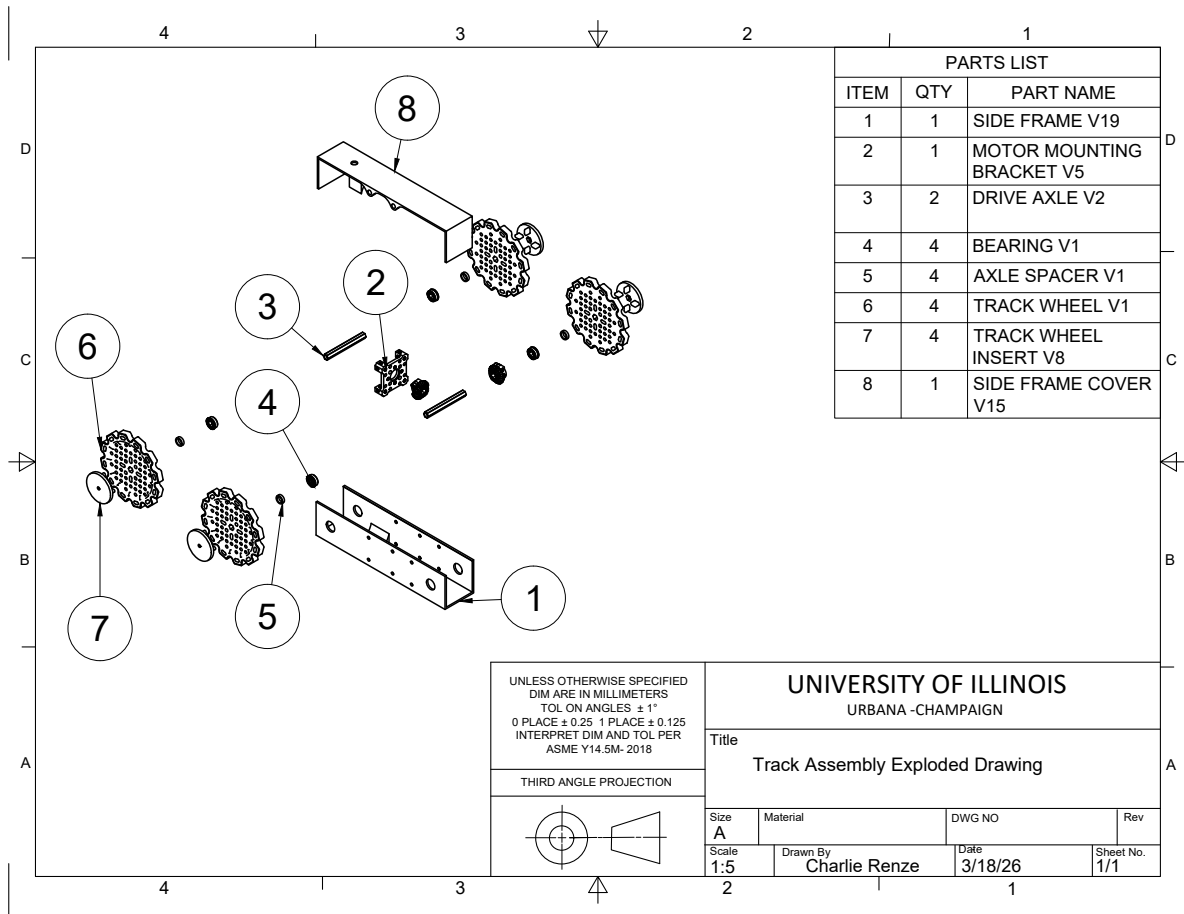


Figure 23: Track assembly exploded drawing

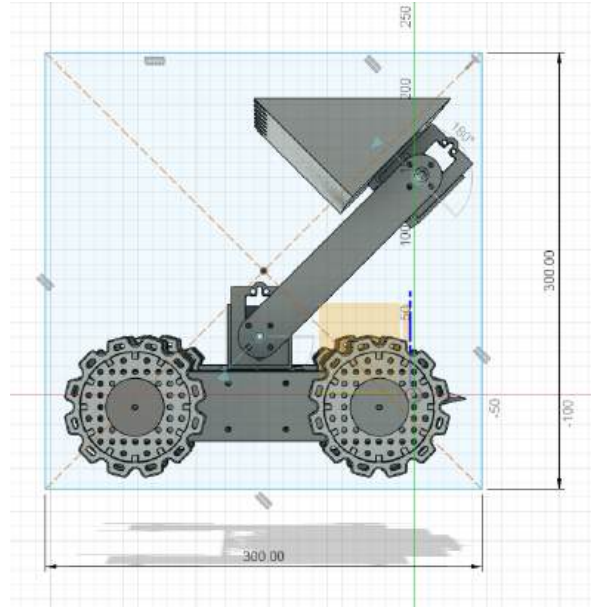


Figure 24: Side profile of R.O.V.E.R with 300 mm x 300 mm sketch

4 Expected Performance and Analysis

The excavation bucket and plough connector were designed to foster efficient resource extraction, transport, and deposit at the resource gathering station while meeting the stringent size requirements. Figure 24 shows a side profile of the R.O.V.E.R with a 300 mm by 300 mm sketch for reference. The figure demonstrates the vehicle is within the size requirements and with sufficient space on the front and rear for track installation.

Using a track wheel diameter of 98.2 mm as shown in Figure 45 and the maximum drive motor speed of 223 RPM, the team calculated the robot could travel at a maximum speed of 1.14 m/s. Given the high friction of the sandy extraterrestrial landscape and the need to conserve battery power, the team reasoned the vehicle could realistically travel at 25% of its theoretical capabilities. This assumption translates to a speed of 0.29 m/s, allowing the R.O.V.E.R to traverse one side of the 2.5 m course in approximately nine seconds. This pace is suitable given the five minute time limit and leaves sufficient time for resource excavation.

The team used the built-in properties tool in Autodesk Fusion to calculate the internal volume of the excavation bucket and resource gathering station. The volume of the bucket is $5.95 \times 10^{-4} \text{ m}^3$ and the volume of the station is $1.3 \times 10^{-2} \text{ m}^3$. These values mean the resource gathering station is approximately 22 times larger than the excavation bucket. Given the time limit, the gathering station is unlikely to overflow and diminish the team's score.

In order to evaluate the servos' capabilities in excavating sand, the team created a sample payload in the excavation bucket with a density of 3000 kg/m^3 . Next, the Fusion's inspect tool was used to calculate the the perpendicular distance from the payload's center of mass to the axis of each servo: 5.6 cm for the 25 kg servo and 24 cm for the 35 kg servo. Given the significantly longer lever arm for the 35 kg motor mounted to the center frame, this unit is the limiting factor in the maximum payload of the excavation bucket. Using a safety factor of 0.8, the team calculated the maximum torque on the motor should be 28 kg and the resulting maximum payload to be 1.2 kg. Then, the size of the payload was adjusted in Fusion to reach this target; Figure 25 shows this setup. The payload occupies approximately two-thirds the total volume of the excavation bucket and is a realistic target to achieve during competition.

Figure 26 shows the R.O.V.E.R in the dumping configuration with a sample resource gathering station. Given the sizing limitations of the vehicle, the bucket does not reach completely over the upper lip of the container. Since the vehicle is already near the bounds of the sizing sketch, the team considered incorporating a power-extending plough connector to raise the height of the bucket when dumping, but reasoned that this mechanism would be too complex and unreliable. Additionally, it

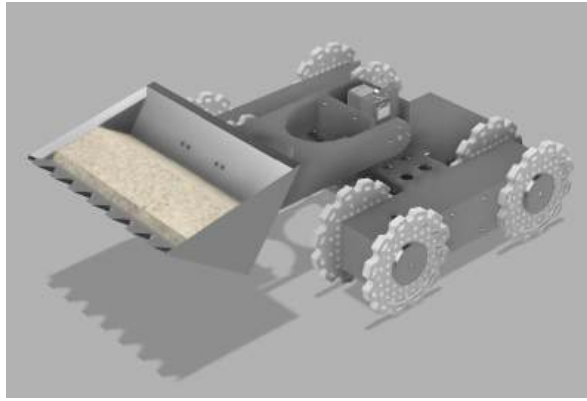


Figure 25: R.O.V.E.R with simulated payload for use in verifying servo capabilities

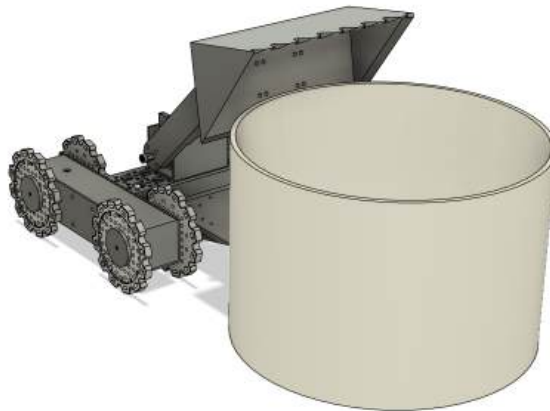


Figure 26: R.O.V.E.R in dumping configuration

would consume additional energy, reducing the power available to other components of the R.O.V.E.R. After discussing these limitations, the team predicted the existing configuration would be sufficient for dumping resources into the resource gathering station.

5 Design for Manufacturing and Assembly (DFMA) Analysis

Throughout the design process, the team implemented DFMA principles to streamline manufacturing and assembly. The overarching design methodology was to harness the unique capabilities of additive manufacturing to consolidate parts, enabling efficient assembly.

The most prominent example of this approach is the R.O.V.E.R's modular design consisting of a center frame, side frames, track assemblies, and servo housings. These components are standardized to improve assembly efficiency, easily configurable to adapt to new design elements, and can be mass-produced to replace broken parts during testing. For instance, the servo housings are identical even though they are mounted in different orientations, eliminating the need for a unique part.

Another example of this philosophy can be seen in the side frames, where the enclosure incorporates structural elements, motor supports, bearing mounting points, and holes for fasteners into a single part. As described in section 2.2, the side frames were initially designed as a single part, but were separated into a main frame and cover to allow for easier assembly. The frames are identical to eliminate confusion during the build process and include projected holes on the outer faces. Students can use these holes to tighten fasteners using the long end of an Allen key, eliminating the tedious process of turning a screw one-quarter revolution at a time. Lastly, the cross-section analysis in Figure 27 shows the semi-spherical drive motor supports used to eliminate additional unique components.

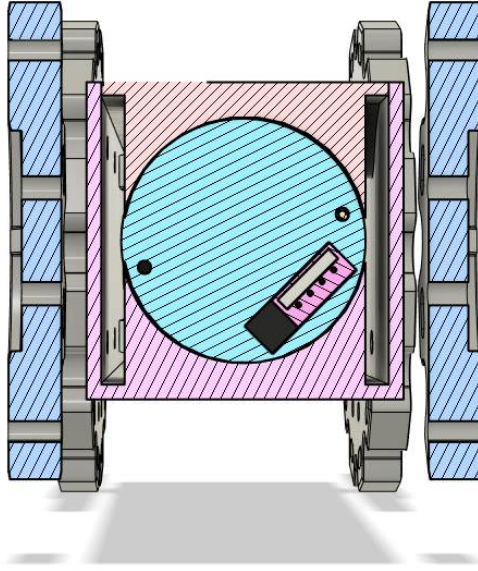


Figure 27: Section analysis with semi-spherical drive motor supports

The team considered combining the entire frame into a single part, but ultimately decided against this approach because it does not allow for incremental design iterations as the vehicle’s development progresses. If the team were to change the design of a component, the entire frame would need to be reprinted and all electronic components removed and reinstalled, undermining any efficiencies gained through part reduction. Additionally, the internal features would require a vast quantity of supports and the part may fail during printing, resulting in unnecessary material waste.

The team streamlined their selection of mechanical fasteners where possible to simplify purchasing and assembly. Standard M4 screws were used in the chassis and servo housings, but specialty M3 screws were required for the servo axles and track links. This compromise was considered acceptable given the unique needs of the frame, axle, and track connections. Finally, the team largely eliminated the use heat set inserts, which are time-consuming to install, through printed thread patterns on Nylon components.

6 Design for Additive Manufacturing (DFAM) Analysis

The team also rigorously implemented DFAM principles throughout the design process to improve assembly efficiency, reduce material waste, and increase part durability.

The majority of the robot’s components were printed from PLA because of its low cost and sustainable production. Throughout the iterative design process, printing parts from PLA significantly reduced material cost and environmental impact compared to petroleum-based plastics. For PLA parts, a tri-hexagonal infill pattern with 20% infill density was used to reduce weight and material usage. Nylon-based SLS printing was used to create the detailed thread patterns on the motor mounting bracket and center frame connector. The complex geometry of the track links was also created using this technology.

The team minimized the use of supports during printing through strategic print orientation and increased the minimum support area from 4 mm² to 7 mm². This change eliminate supports on fastener holes and significantly reduced post-processing. Fastener holes for M4 screws were designed slightly larger than necessary to allow for misalignment. A consistent wall thickness of 2 mm in frames and 4 mm in brackets was used to mitigate warping, boost print reliability, and improve the surface finish.

7 Testing

The team conducted iterative testing to measure the performance of the R.O.V.E.R and ensure it functioned as intended. At the time of writing, some testing has been accomplished, but a full test program could not be realized due to time constraints. The following sections describe the completed and planned tests.

7.1 Completed Tests

After assembling the robot, the team began with basic tests of the vehicle's functionality to ensure the design would work as intended at the competition. The following testing steps were taken:

1. Connected battery terminals to power supply to ensure electrical components powered on
2. Paired xBox controller via Bluetooth and verified the drive and servo motors responded to user inputs
 - (a) The reaction time of motor control for mechanism movement was imperceptible to the operator
3. Drove R.O.V.E.R on tabletop to ensure track functionality
4. Confirmed stable Bluetooth connection between R.O.V.E.R and xBox controller at a distance of fifty feet
5. Verified stable connection of FPV cameras to control laptop via Wi-Fi as shown in Figure 29
6. Tested wide angle FPV lenses to ensure adequate forward and rearward visibility
7. Drove R.O.V.E.R on student-built sand pit, shown in Figures 30 and 28, to replicate competition environment
 - (a) The team began testing with a tethered connection to the power supply to ensure sufficient electrical power and longer runtime, then later transitioned to battery power.
 - (b) After several minutes of testing, the team found the original SLS-printed nylon bevel gears shown in Figure 31 were significantly worn. Even though it slightly reduced the team's additive manufacturing score, the team elected to replace the nylon gears with identical steel models to ensure a reliable drive mechanism.

With the exception of the failed nylon bevel gears, these initial tests verified that the R.O.V.E.R functions as intended. The track system allows the vehicle to efficiently navigate the uneven competition landscape and the xBox controller provided sufficient control of the drive and servo motors. Given the limited scope of these trials, no design changes were made.

7.2 Planned Tests

The following tests are planned to more comprehensively evaluate the performance of the R.O.V.E.R. The team will make corresponding adjustments to the vehicle's design and excavation strategy to ensure a successful competition run.

1. Measure speed of R.O.V.E.R and compare to expectations listed in Section 4
 - (a) If the vehicle travels as fast or faster than expected, the team will attempt to navigate the entire landscape to search for high-value metal resources. However, if the robot is slower than anticipated, the team will excavate resources close to the resource gathering station to ensure the greatest mass is deposited at the end of the competition time.
2. Measure battery life of robot during test competition run and ensure it exceeds the five minute time limit
3. Ensure no sand or dust intrusion in the side frames



Figure 28: Testing the R.O.V.E.R on the sand pit



Figure 29: Students testing connection of FPV Cameras using Wi-Fi



Figure 30: Student-built sand pit designed to replicate competition environment



Figure 31: Original nylon bevel gears; later replaced with identical steel models

- (a) If sand is detected, the team will add a gasket along the perimeter of the frames to protect the drive motors
- 4. Drive vehicle on sand pit with scoop pushing and lifting sand to replicate what will actually occur in competition
 - (a) The vehicle should be able to lift and maneuver the excavation bucket when it is half full to ensure sufficient resources can be deposited in the gathering station.
- 5. Test ability of robot to retrieve object in known location buried under 10 cm of sand
- 6. Test whether R.O.V.E.R can empty excavation bucket into resource gathering station
 - (a) If the current design is too short, the team will redesign the plough connector to ensure the excavation bucket can reach fully over the upper lip of the resource gathering station. The entire competition hinges on the R.O.V.E.R's ability to perform in this regard, so the team will focus significant attention on this test.
- 7. Measure time required to execute a full digging motion from resource identification to deposit in resource gathering station
 - (a) The team hopes that this process will take less than 40 seconds, allowing for five complete excavation and dumping motions.
- 8. Confirm ability of FPV camera to connect to control laptop using Wi-Fi from a distance of fifty feet
- 9. Test ability of metal detector system to detect ferrous material from 10 cm away

8 Conclusion

The R.O.V.E.R successfully integrates creative design thinking, additive manufacturing principles, and electronics integration to efficiently extract resources from an extraterrestrial environment. Through an iterative design process, the team evaluated multiple concepts for propulsion, chassis, and excavation mechanisms, ultimately converging on a robust and efficient solution that meets the competition's time and dimensional constraints.

A tracked propulsion system was selected to maximize traction and reliability on uneven terrain, while a modular chassis design enabled rapid prototyping and flexible integration of subsystems. The excavation bucket emerged as the optimal digging mechanism through systematic evaluation using a Pugh matrix, offering a balance of simplicity, reliability, and performance. The final design minimizes complexity by eliminating the need for an auxiliary conveyor belt, reducing both assembly time and potential failure points.

Design for Manufacturing and Assembly (DFMA) and Design for Additive Manufacturing (DFAM) principles were central to the development of the R.O.V.E.R. The team leveraged additive manufacturing to consolidate components, integrate thread patterns, and simplify assembly while maintaining structural integrity. Strategic decisions such as consistent wall thickness and optimized print orientation improved manufacturability and reduced post-processing requirements. The resulting design achieves an additive manufacturing utilization of 87.8%, reflecting a strong alignment with competition objectives.

Preliminary testing verified the functionality of the propulsion and control systems. Planned testing will further validate system performance, including battery life, excavation efficiency, and speed under competition conditions. These efforts will guide final refinements to the excavation mechanism to ensure optimal operation during the competition.

Overall, the R.O.V.E.R represents a well-balanced engineering solution that prioritizes reliability, manufacturability, and performance. The design effectively utilizes additive manufacturing technologies and positions the team for a competitive performance at the ASME IAM3D competition.

9 Appendix: Part Drawings

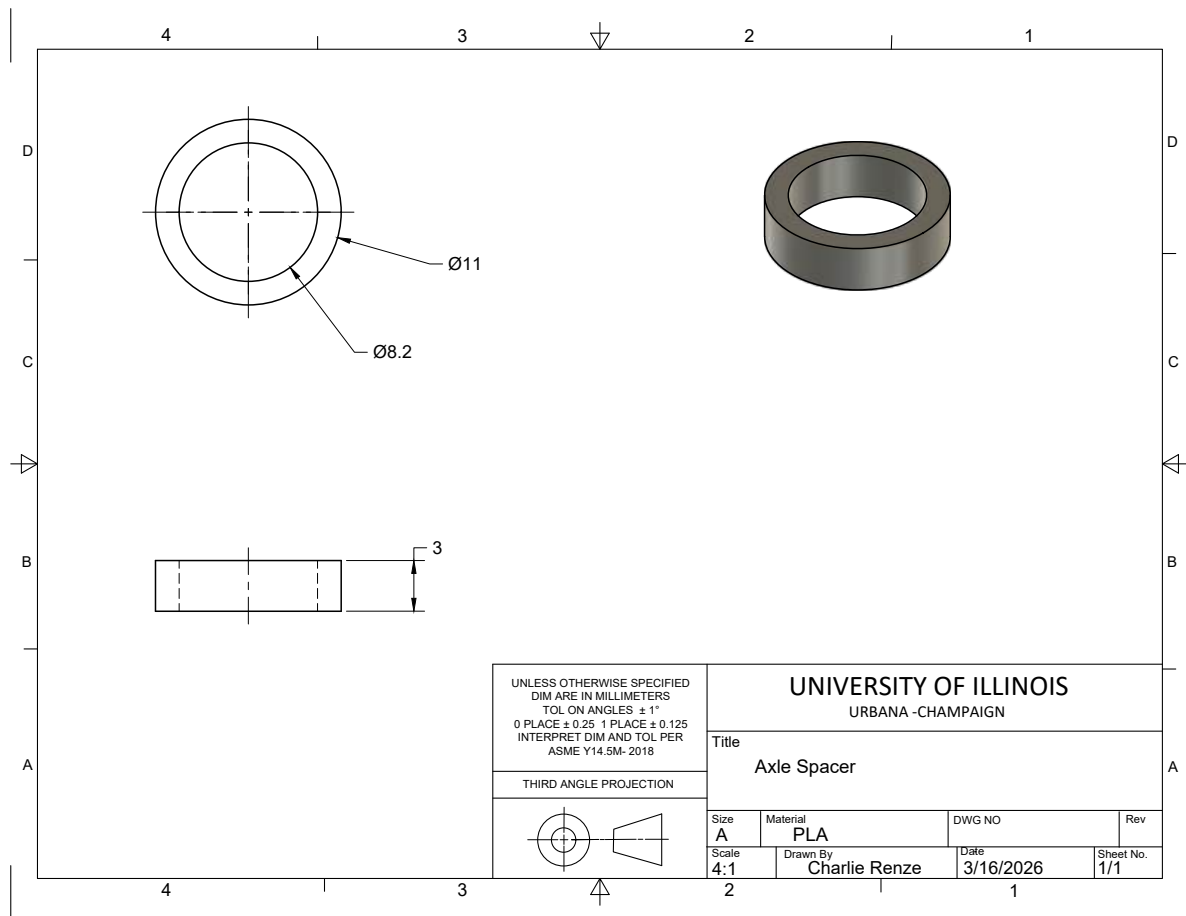


Figure 32: Axle spacer drawing

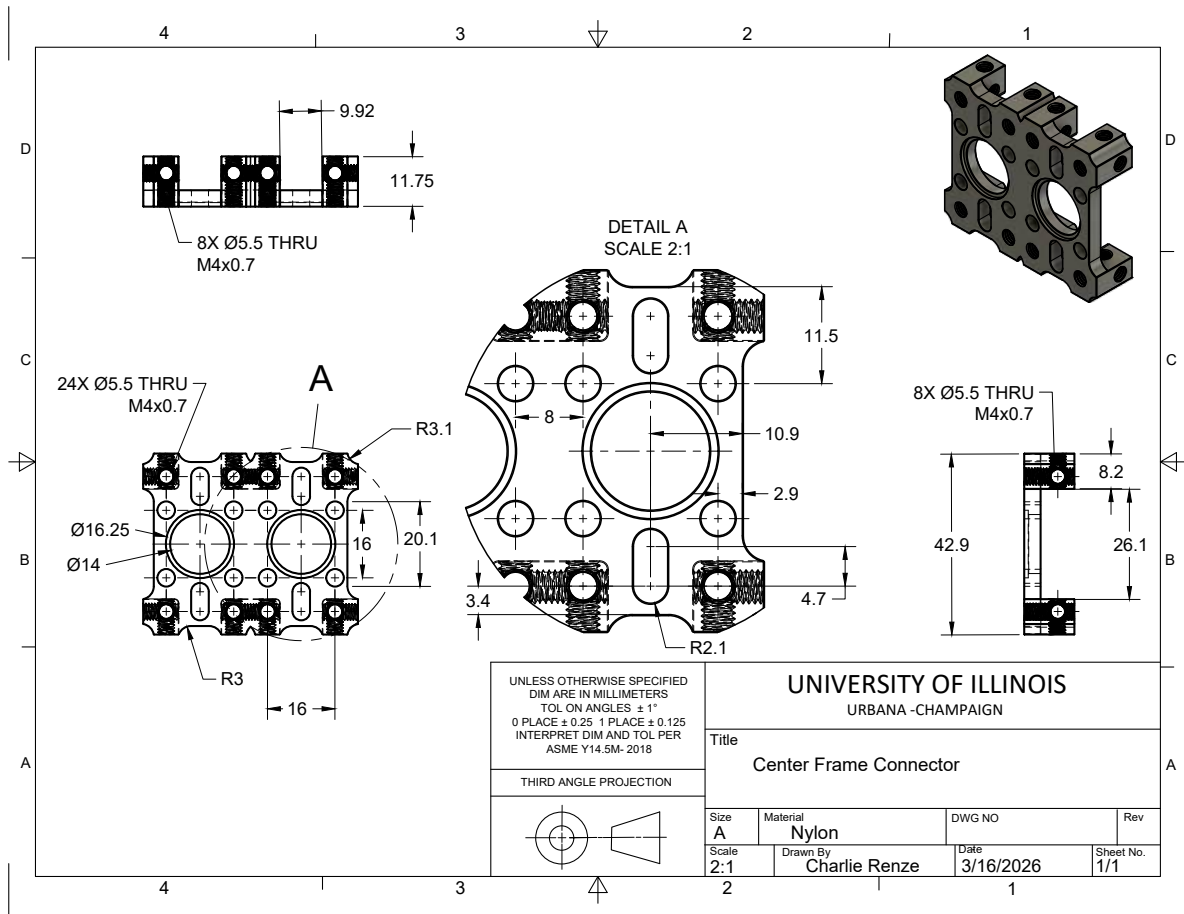


Figure 33: Center frame connector drawing

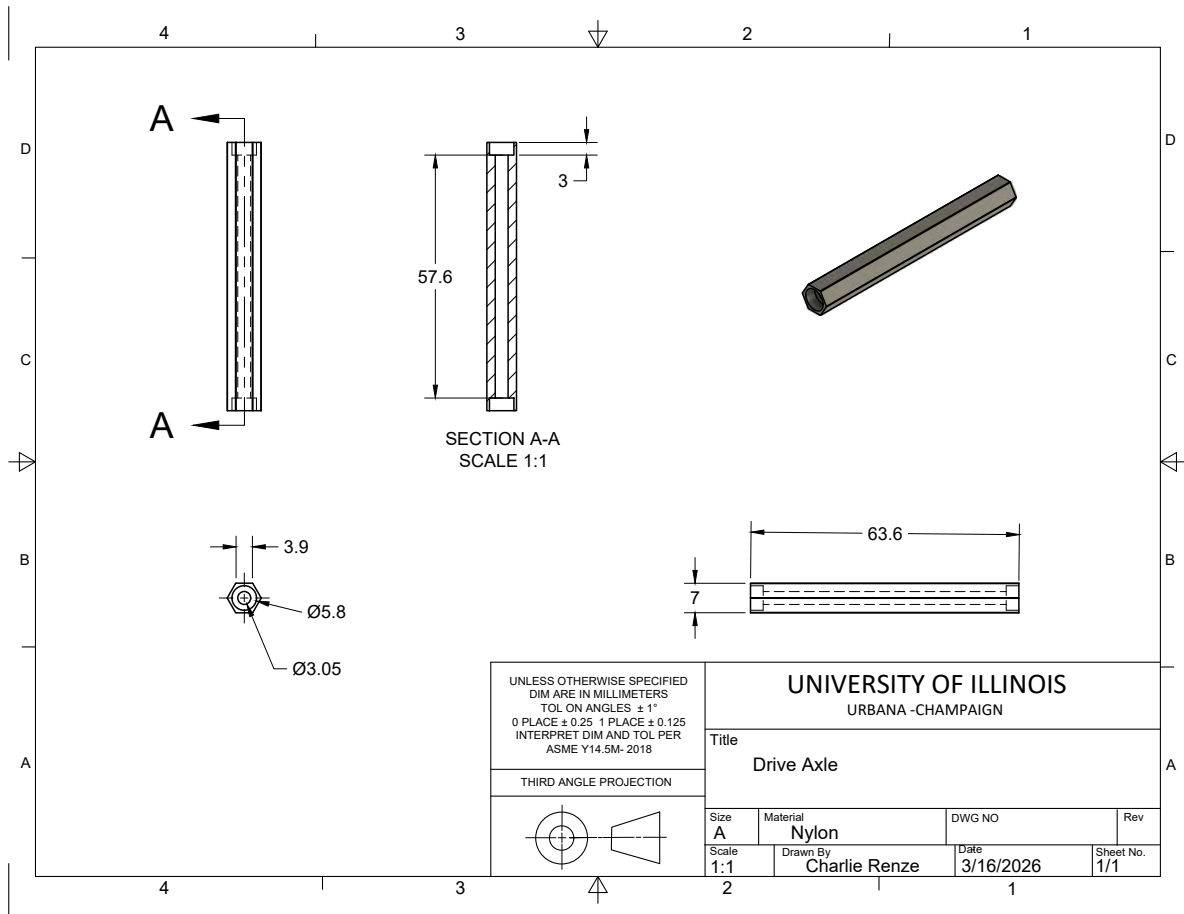


Figure 35: Drive axle drawing

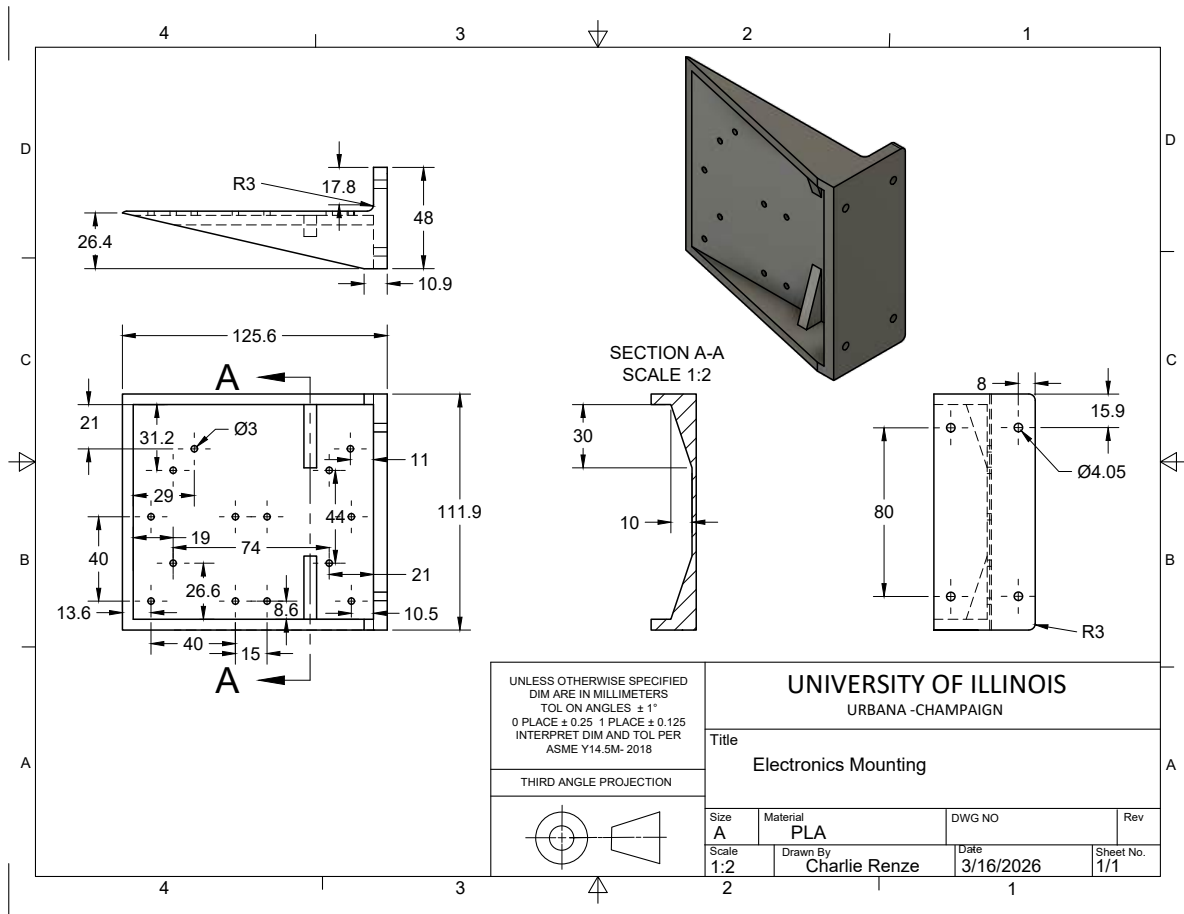


Figure 36: Electronics mounting drawing

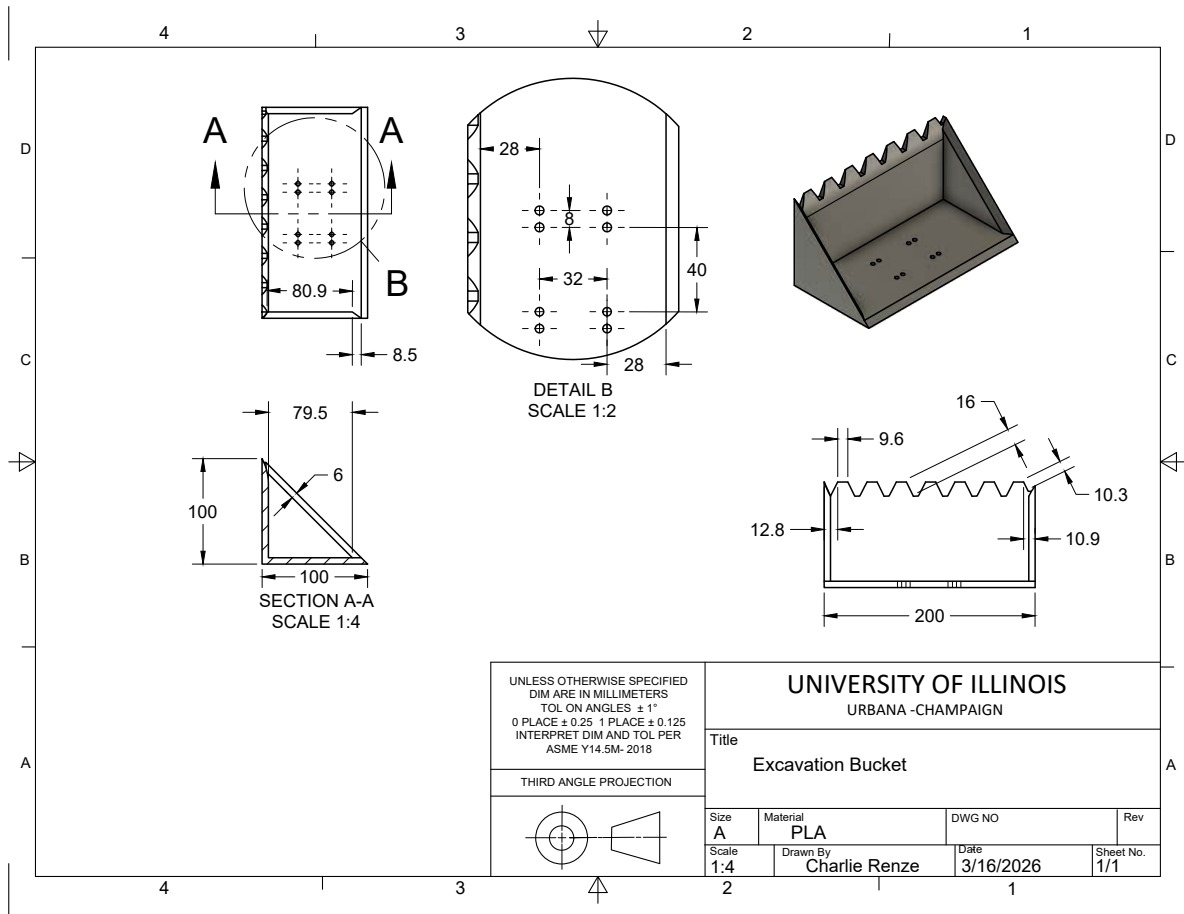


Figure 37: Excavation bucket drawing

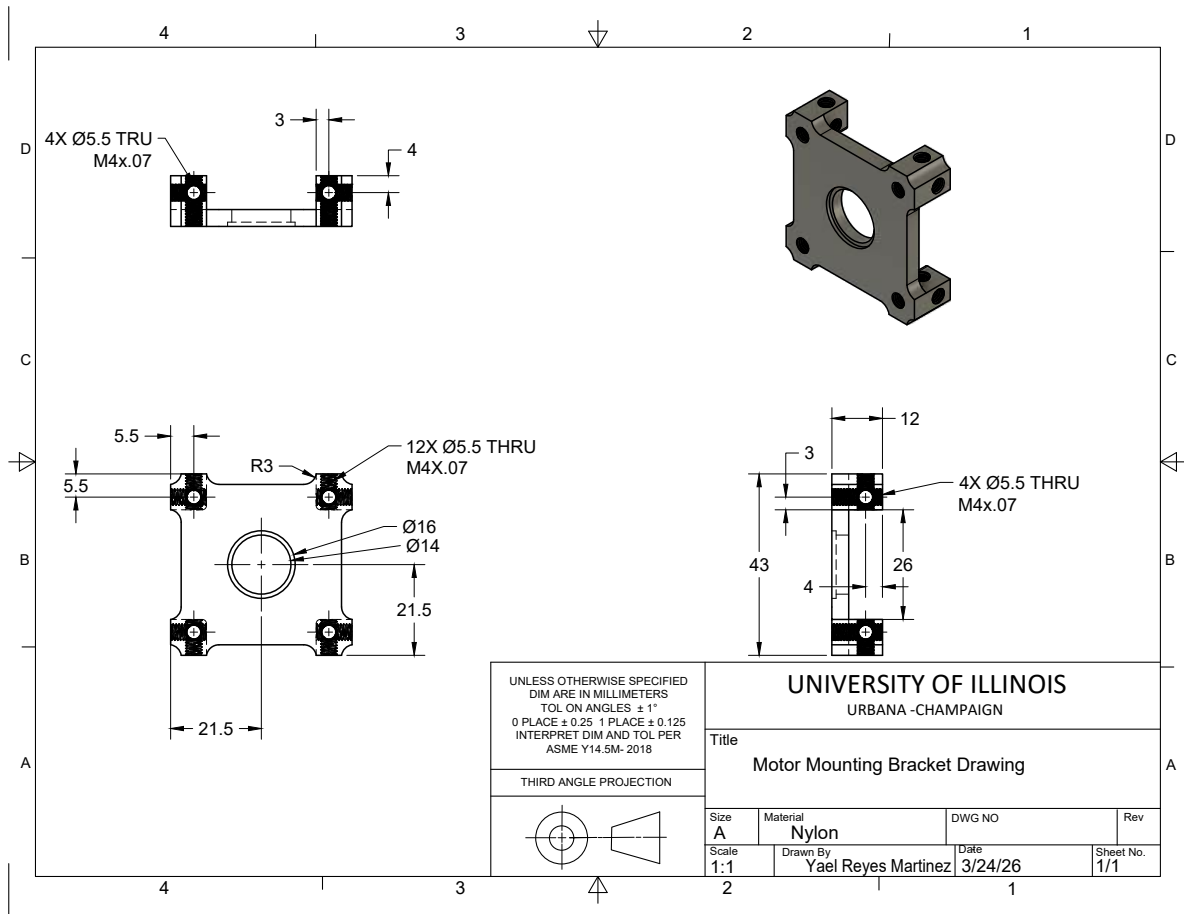


Figure 38: Motor mounting bracket drawing

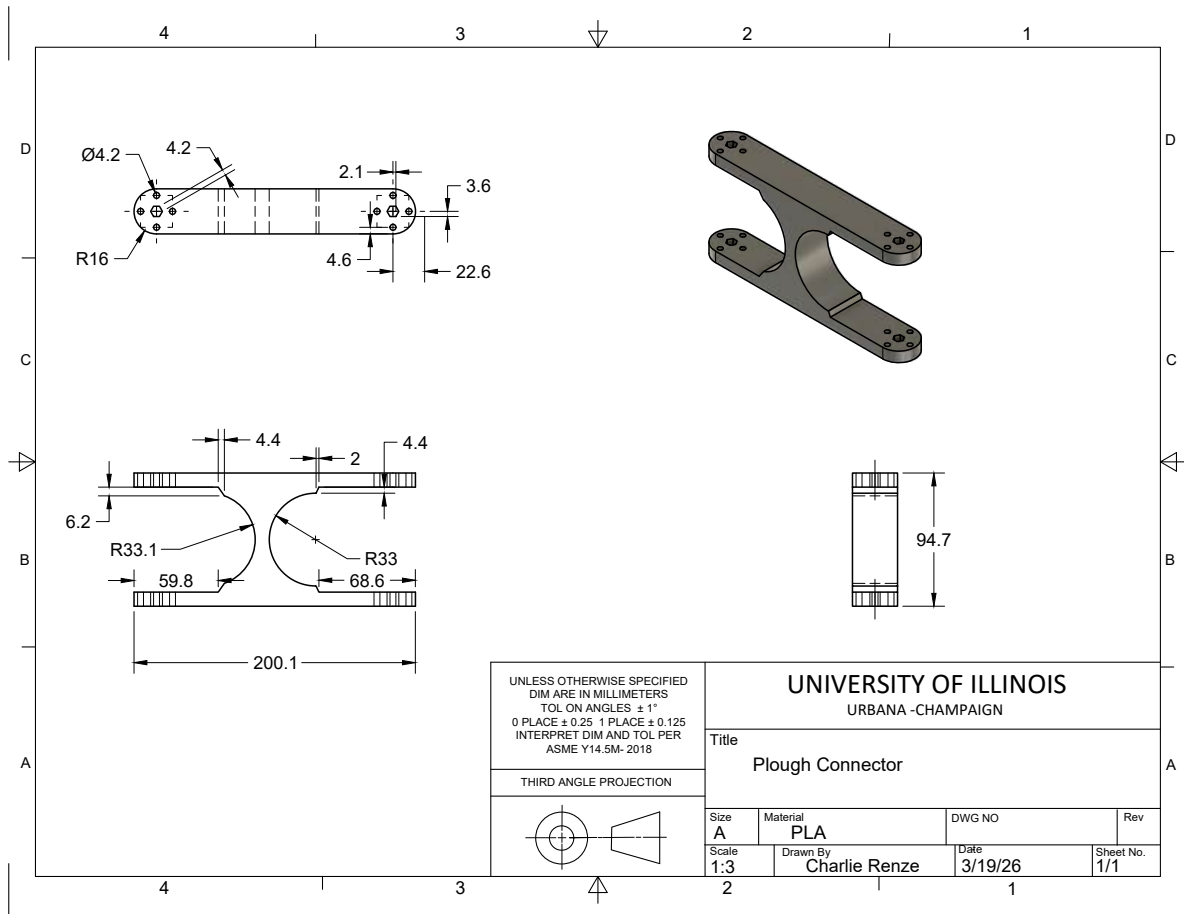


Figure 39: Plough connector drawing

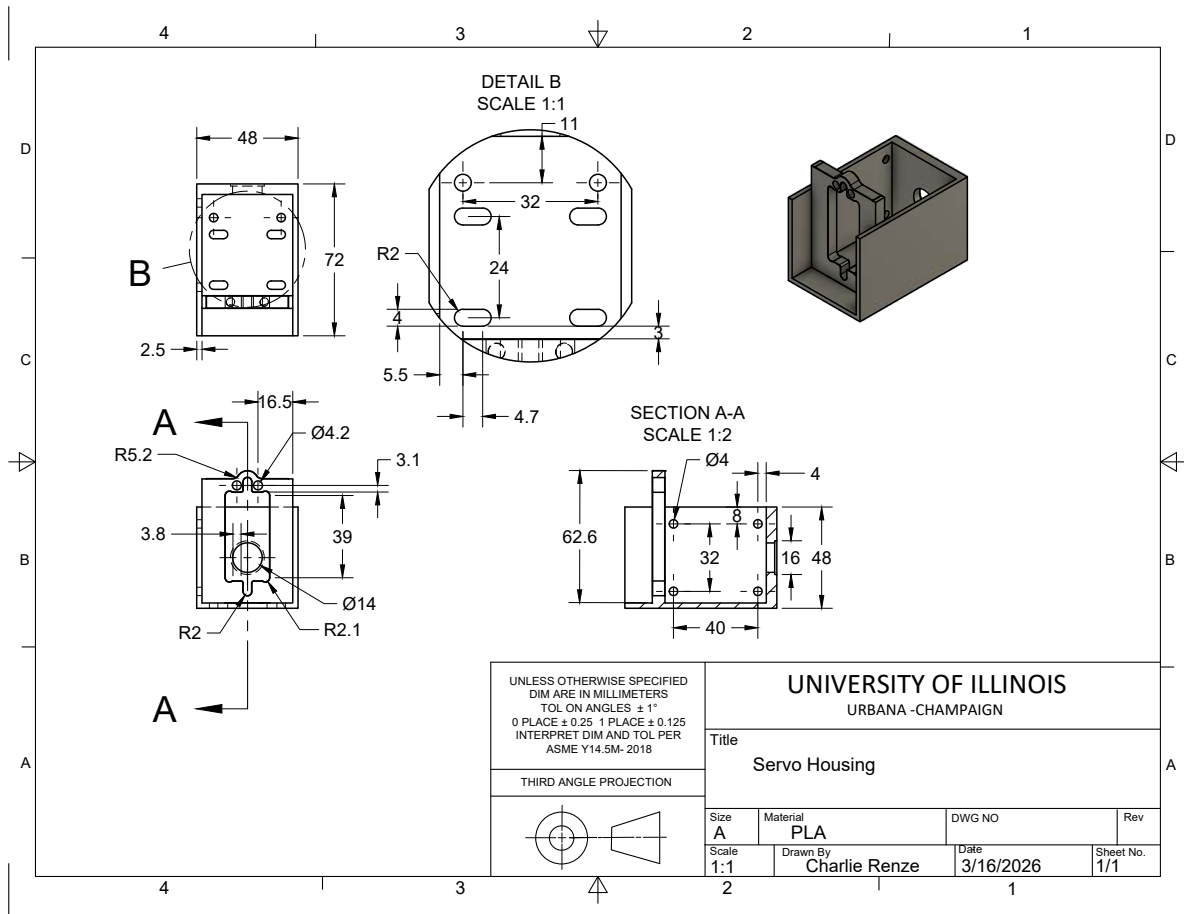


Figure 40: Servo housing drawing

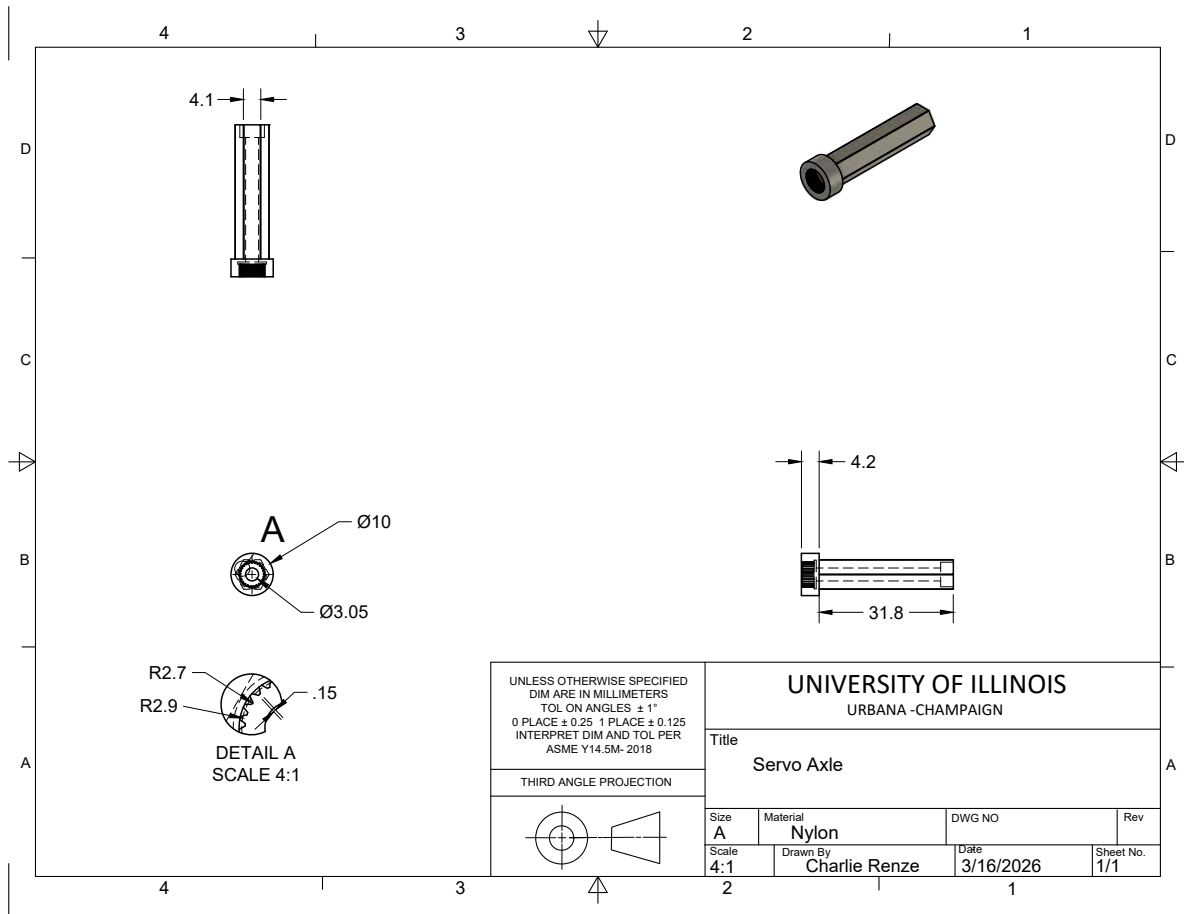


Figure 41: Servo axle drawing

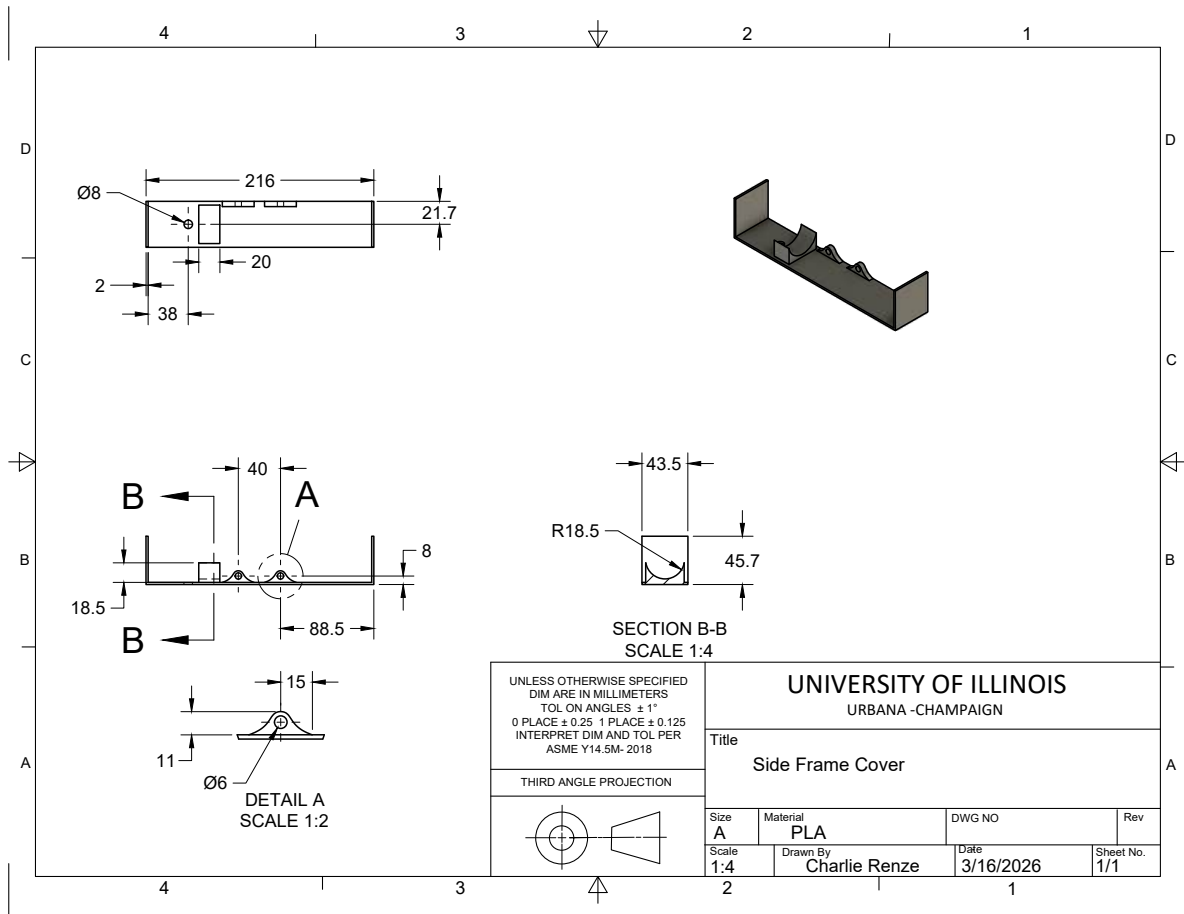


Figure 42: Side frame cover drawing

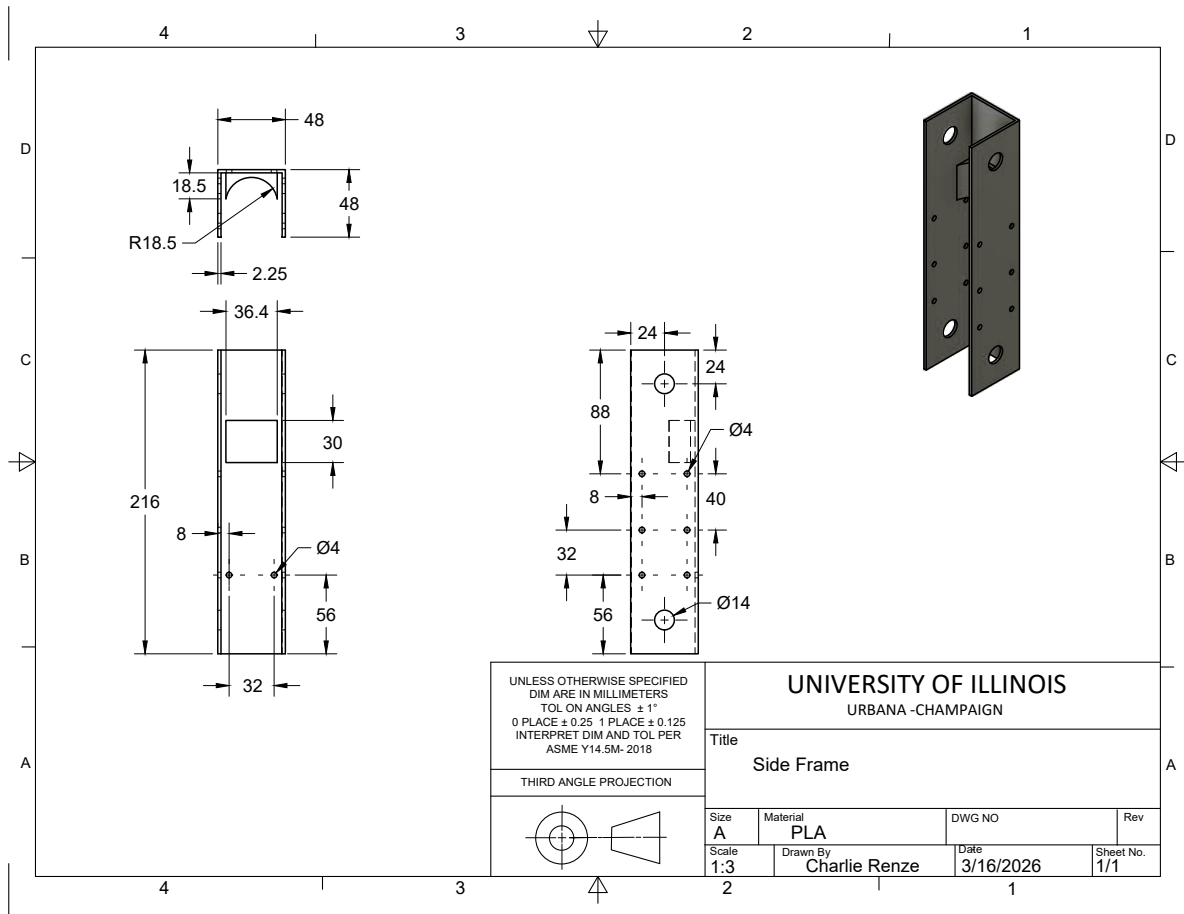


Figure 43: Side frame drawing

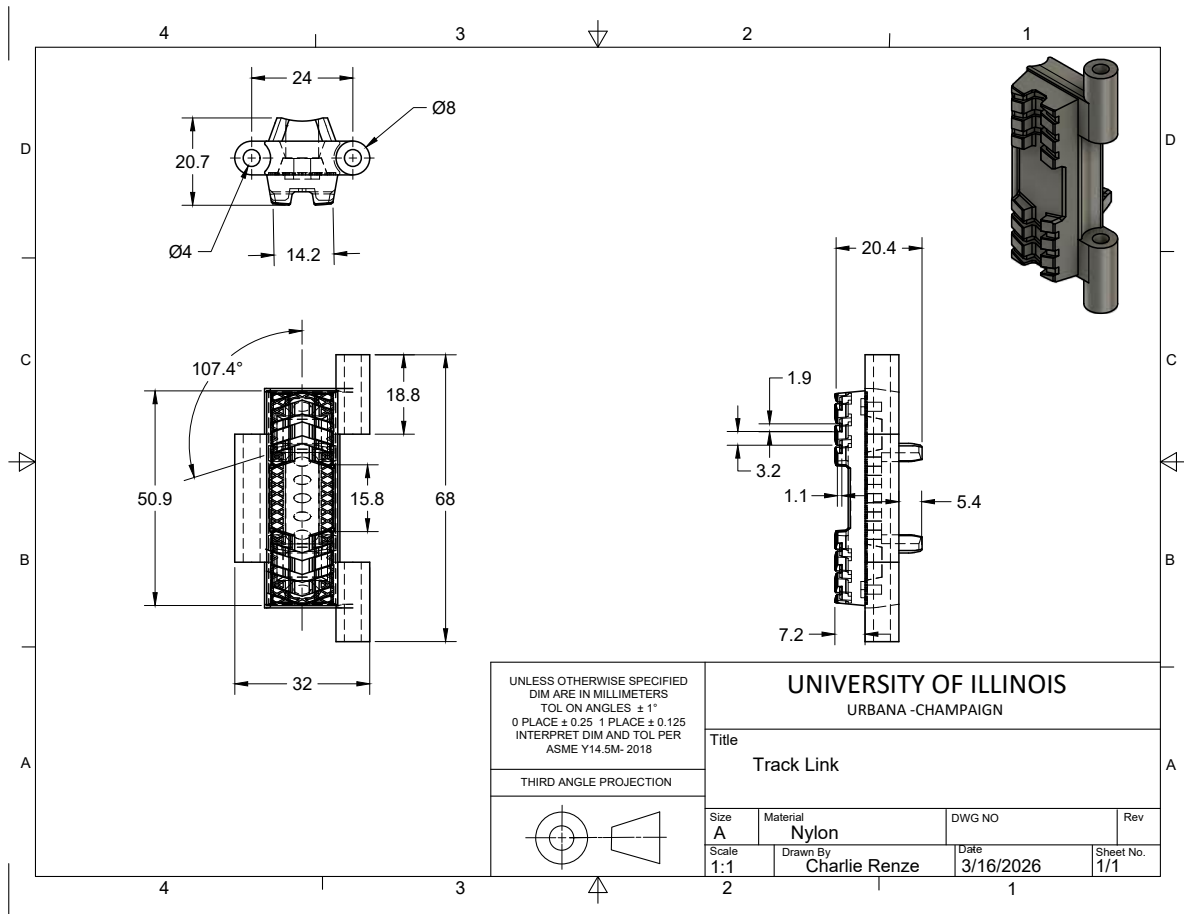


Figure 44: Track link drawing

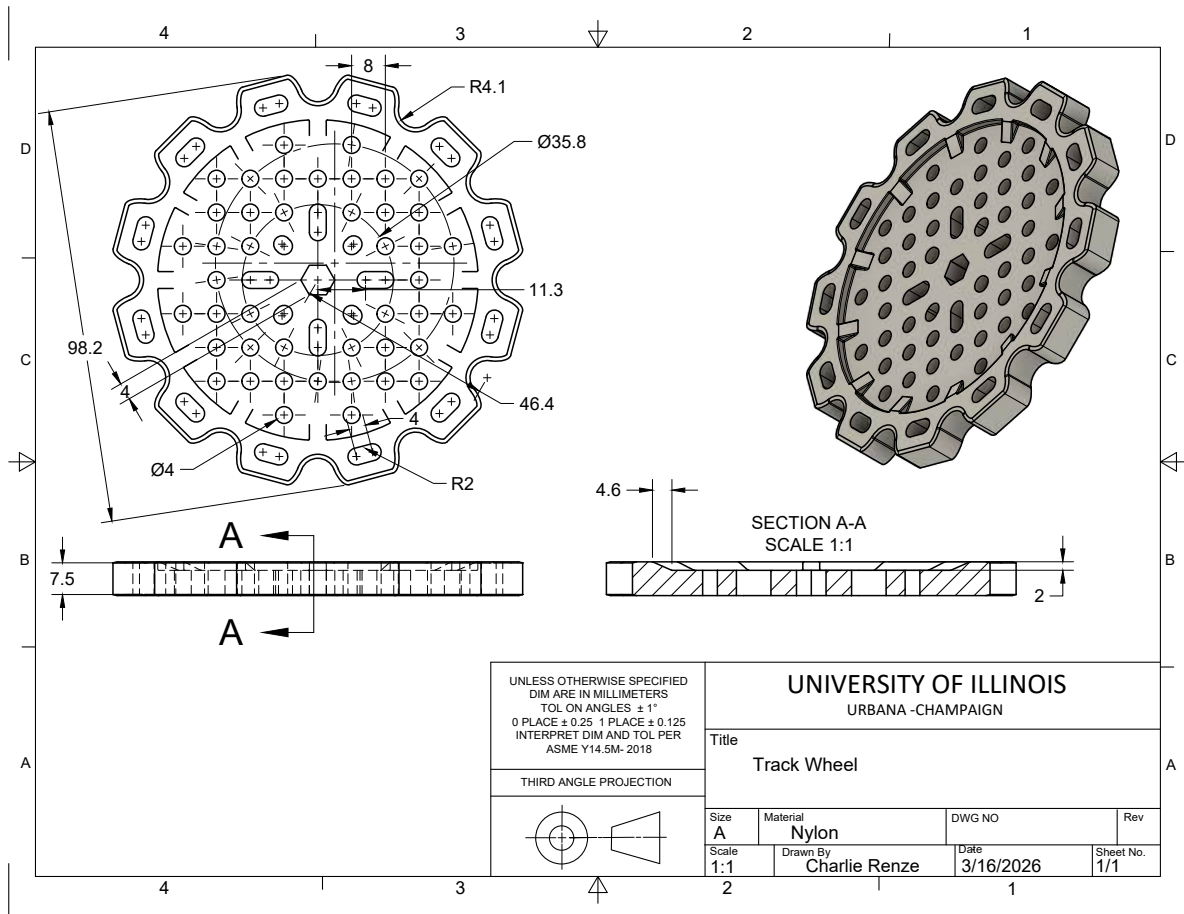


Figure 45: Track wheel drawing

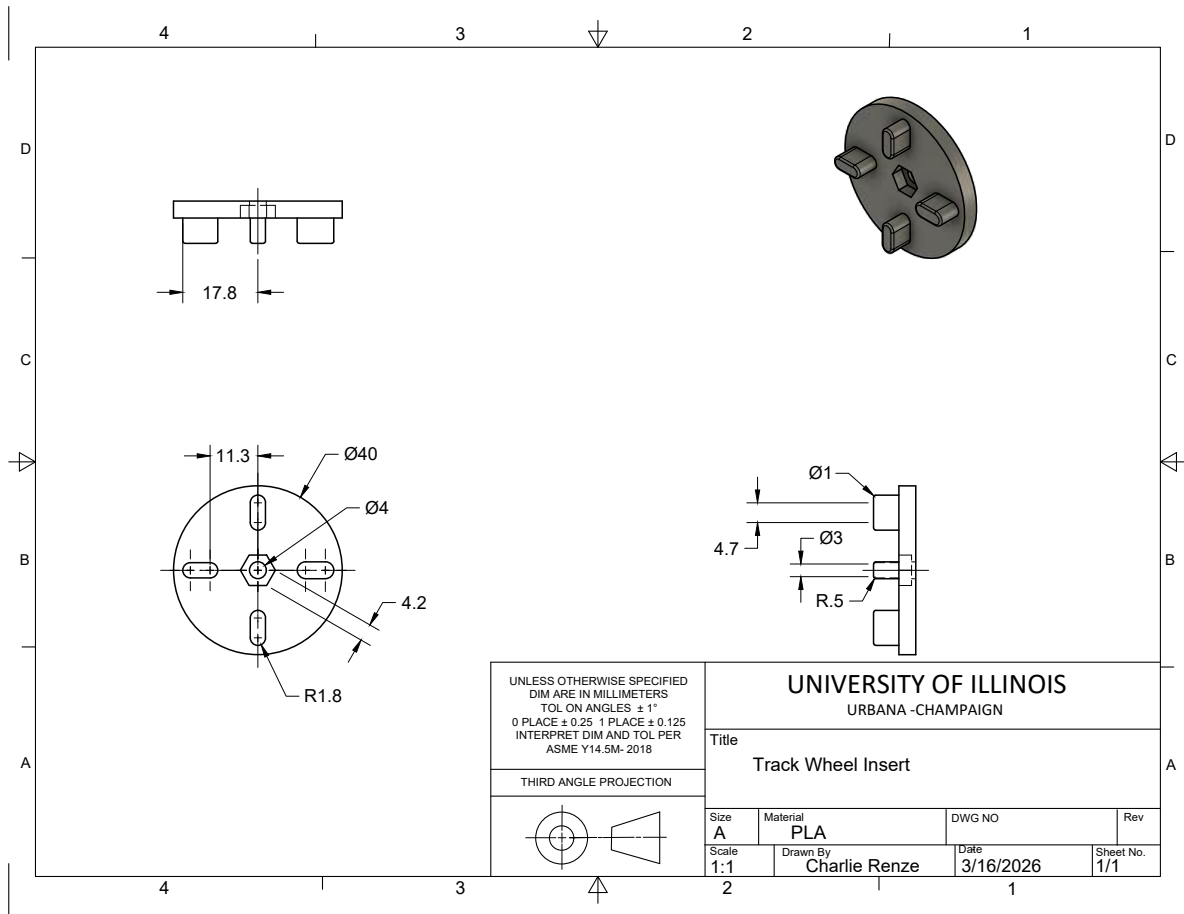


Figure 46: Track wheel insert drawing